

The Journal

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Editorial

by Rob J Hyndman

Editorial changes

This is the last issue for 2025, and so it also marks the end of Mark van der Loo's term as an Executive Editor of the *R Journal*. We thank him for his work over the last four years, and especially for his service as Editor-in-Chief during 2024. He has consistently worked to improve the *R Journal*, and raise the standard of published articles and packages, and we are grateful for his contributions.

This is also the last issue for me as Editor-in-Chief of the *R Journal*. It has been an honour and a pleasure to serve in this role, and a privilege to work with a great group of editors and associate editors. I am happy to hand the reins to Dr Emi Tanaka as the incoming Editor-in-Chief, who has been an Executive Editor since 2024. Emi is a Senior Lecturer at the Australian National University, and is a very active member of the R community with several contributed packages on CRAN.

We also welcome Professor Vincent Arel-Bundock, from the University of Montreal, as a new Executive Editor for the period 2026–2029. He joins Emi Tanaka, Emily Zabor and me as the team of Executive Editors for 2026.

We also welcome two new Associate Editors: Selçuk Korkmaz and Maciej Beręsewicz. We are grateful to them, and the large team of Associate Editors, for their willingness to contribute to the *R Journal*.

Finally, thanks to Mitchell O'Hara-Wild, who is stepping down as Technical Editor of the Journal. He has provided wonderful support to the *R Journal* over many years, solving countless bewildering technical issues in order to make the Journal website function smoothly. In his place, we welcome Abhishek Ulayil, who will be the new Technical Editor from 2026. We are grateful to Abhishek for taking on this valuable role.

New guidelines

We recently introduced new guidelines for papers about R packages (which covers the vast majority of the papers we publish). The new guidelines are available on the [R Journal website](https://journal.r-project.org). The intention is to make the expectations for authors clearer, and to improve the quality and consistency of articles published in the *R Journal*. Prospective authors should follow these guidelines when preparing their submissions.

In this issue

On behalf of the editorial board, I am pleased to present Volume 17 Issue 4 of the *R Journal*. This issue features 15 research articles. Each article relates to an R package available on CRAN, providing an overview of the package, its functionality, and examples of its use. Supplementary material for each article, with fully reproducible code, is available for download from the Journal website. We also include news from CRAN, Bioconductor, the Forwards Taskforce, and the R Foundation.

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csurvey: Implementing Order Constraints in Survey Data Analysis

by Xiyue Liao and Mary C. Meyer

Abstract Recent work in survey domain estimation has shown that incorporating a priori assumptions about orderings of population domain means reduces the variance of the estimators. The new R package `csurvey` allows users to implement order constraints using a design specified in the well-known survey package. The order constraints not only give estimates that satisfy a priori assumptions, but they also provide smaller confidence intervals with good coverage, and allow for design-based estimation in small-sample or empty domains. A test for constant versus increasing domain means is available in the package, with generalizations to other one-sided tests. A cone information criterion may be used to provide evidence that the order constraints are valid. Examples with well-known survey data sets show the utility of the methods. This package is now available from the Comprehensive R Archive Network at <http://CRAN.R-project.org/package=csurvey>.

1 Introduction

We assume that a finite population is partitioned into a number of domains, and the goal is to estimate the population domain means for the study variable and provide valid inference, such as confidence intervals and hypothesis tests.

Order constraints are a common type of *a priori* knowledge in survey data analysis. For example, we might know that salary increases with job rank within job type and location, or average cholesterol in a population increases with age category, or test scores decrease as poverty increases, or that the amount of pollution decreases with distance from the source. There might be a “block ordering” of salaries by location, where the locations are partitioned into blocks where salaries for all locations in one block (such as major metropolitan areas) are assumed to be higher, on average, than salaries in another block (such as rural areas), without imposing orderings within the blocks.

The order constraints may be imposed on domain mean estimates, or systematic component estimates if the response variable is not Gaussian but belongs to an exponential family of distributions. For example, if we are estimating the proportion of people with diabetes in a population, the study variable might be binary (whether or not the subject has diabetes), and perhaps we want to assume that diabetes prevalence increases with age, within ethnicity and socio-economic categories. The `csurvey` package extends the `survey` (Lumley, 2004) package in that it allows the user to impose linear inequality constraints on the domain means. If the inequality constraints are valid, this leads to improved inference, as well as the ability to estimate means for empty or small-sample domains without additional assumptions. The `csurvey` package provides constrained estimates of means or proportions, estimated variances of the estimates, and confidence intervals where the upper and lower bounds of the intervals also satisfy the order constraints.

The `csurvey` package implements one-sided tests. Suppose the study variable is salary, and interest is in whether the subject’s salary is affected by the level of education of the subject’s father, controlling for the subject’s education level and field. The null hypothesis is that there is no difference in salary by the level of father’s education. The one-sided test with alternative hypothesis that the salary increases with father’s education has a higher power than the two-sided test with the larger alternative “salary is not the same across the levels of father’s education.”

Finally, the `csurvey` package includes graphical functions for visualizing the order-constrained estimator and comparing it to the unconstrained estimator. Confidence bands can also be displayed to illustrate estimation uncertainty.

This package relies on **coneproj** (Liao and Meyer, 2014) and **survey** for its core computations and handling of complex sampling designs. The domain grid is constructed using the **data.table** (Barrett et al., 2025) package. Additional functionality, such as data filtering, variable transformation, and visualization of model results, leverages functions from **igraph** (Csárdi et al., 2025), **dplyr** (Wickham et al., 2023), and **ggplot2** (Wickham, 2016). When simulating the mixture covariance matrix and the sampling distribution of the one-sided test statistic, selected functions from **MASS** (Venables and Ripley, 2002) and **Matrix** (Bates et al., 2025) are employed.

2 Research incorporated in csurvey

The **csurvey** package provides estimation and inference on population domain variables with order constraints, using recently developed methodology. Wu et al. (2016) considered a complete ordering on a sequence of domain means. They applied the pooled adjacent violators algorithm Brunk (1958) for domain mean estimation, and derived asymptotic confidence intervals that have smaller width without sacrificing coverage, compared to the estimators that do not consider the ordering. Oliva-Avilés et al. (2020) developed methodology for partial orderings and more general constraints on domains. These include block orderings and orderings on domains arranged in grids by multiple variables of interest. Xu et al. (2021) refined these methods by proposing variance estimators based on a mixture of covariance matrices, and showed that the mixture covariance estimator improves coverage of confidence intervals while retaining smaller interval lengths. Liao et al. (2024) showed how to use the order constraints to provide conservative design-based inference in domains with small sample sizes, or even in empty domains, without additional model assumptions. Xu and Meyer (2023) developed a test for constant versus increasing domain means, and extended it to one-sided tests for more general orderings. Oliva-Avilés et al. (2019) proposed a cone information criterion (CIC) for survey data as a diagnostic method to measure possible departures from the assumed ordering. This criterion is similar to Akaike's information criterion (AIC) (Akaike, 1973) and the Bayesian information criterion (BIC) (Schwarz, 1978), and can be used for model selection. For example, we can use CIC to choose between an unconstrained estimator and an order constrained estimator. The one with a smaller CIC value will be chosen.

All of the above papers include extensive simulations showing that the methods substantially improve inference when the order restrictions are valid. These methods are easy for users to implement in **csurvey**, with commands to impose common orderings.

3 How to use csurvey

Statisticians and practitioners working with survey data are familiar with the R package **survey** (see Lumley (2004)) for analysis of complex survey data. Commands in the package allow users to specify the survey design with which their data were collected, then obtain estimation and inference for population variables of interest. The new **csurvey** package extends the utility of **survey** by allowing users to implement order constraints on domains.

The **csurvey** package relies on the functions in the **survey** package such as `svydesign` and `svrepdesign`, which allow users to specify the survey design. This function produces an object that contains information about the sampling design, allowing the user to specify strata, sampling weights, etc. This object is used in statistical functions in the **csurvey** package in the same manner as for the **survey** package. In addition, the mixture covariance matrix in Xu et al. (2021) is constructed from an initial estimate of covariance obtained from **survey**.

Consider a finite population with labels $U = \{1, \dots, N\}$, partitioned into domains U_d , $d = 1, \dots, D$, where U_d has N_d elements. For a study variable y , suppose interest is in